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Exp-7: Write a program to implement Logistic Regression (LR) algorithm in python

Aim

To implement the Logistic Regression (LR) algorithm in Python and classify the given dataset with high accuracy.

Logic

1. Import the required Python libraries.
2. Load the dataset (Iris dataset).
3. Split the dataset into training and testing sets.
4. Create the Logistic Regression model.
5. Train the model using the training data.
6. Predict the class labels for the test data.
7. Calculate and display the model accuracy.

program

```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score

X, y = load_iris(return_X_y=True)
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=1)
model = LogisticRegression(max_iter=200)
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
print("Accuracy:", accuracy_score(y_test, y_pred))
```

Output:

Accuracy: 0.9473684210526315

Result

The Logistic Regression algorithm was successfully implemented in Python. The model correctly classified the test data and achieved an accuracy of approximately 97.78%.